

Hyeonji Shin

ROBOTICS · PERCEPTION & LOCALIZATION

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Education

GIST (Gwangju Institute of Science and Technology) @MPIL M.S. in Mechanical and Robotics Engineering (Adviser: Prof. Pyojin Kim)	Gwangju, South Korea Mar. 2025 - Current
University of Seoul B.S. in Electrical and Computer Engineering	Seoul, South Korea Mar. 2021 - Feb. 2025

Work Experience

Research Internship @UOS Robotics Lab (Adviser: Prof. Myunjoong Hwang) - Optimal Path Planning in Frenet Frame for Autonomous Vehicle Racing Competition - LiDAR Inertial Odometry Comparison in CARLA Simulator - Implementation of 3D Simulator for Autonomous Robot Race Challenge (OSM, CARLA)	Jul. 2023 - Aug. 2024 (Mar. - Aug. 2024) (Mar. - May 2024) (Dec. 2023 - Feb. 2024)
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Teaching Experience

Teaching Assistant of (AI4600/MC4237) Robot Kinematics & Vision @GIST - Conducted a lab study group on feed-forward 3D vision methods (MAST3R-SLAM, VGGT-SLAM2, SLAM3R, MapAnything) - Instructed undergraduate students on MapAnything (3DV 2026) setup and experimental workflow for PBL	Spring 2026
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Publications

International Conference

1. “ThermalManhattan: Robust Thermal Compass under Radiometric Flatness in Manhattan World” Hyeonji Shin, Junpyo Kim, Pyojin Kim	Submitted
2. “PI-EAT-SLAM: Physics-Informed Edge-Aware Thermal Gaussian Splatting SLAM” Junpyo Kim, Seonwook Yeom, Seonmo Yang, Jungil Ham, Hyeonji Shin, Changhyeon Kim, Hyeonbeom Lee, Pyojin Kim	IROS 2026
3. (Edge Track) “SAT-SLAM: Structure-Aware Thermal SLAM Leveraging Structure Regularity of Lines”	3DV 2025

Domestic Conference

1. “Local Path Planning for Dynamic Obstacle Avoidance and High Speed in Autonomous Racing” Seongbin Jo, Hyeonji Shin, Kyuhwan Choi, Taegyeom Lee, Kanghyeon Choi, Dongwook Moon, Myunjoong Hwang	KROS 2025
2. “Comparison of LiDAR-Inertial Odometry for GNSS-denied Areas in Autonomous Vehicle” Hyeonji Shin, UiHun Sagong, Myunjoong Hwang	ICROS 2024

Projects

Spring 2026	Fine-tuning and Distillation for Campus-scale Visual Localization Main Role — Proposal, Dataset curation, Fine-tuning of OrienterNet	CV (AI5303) Term Proj.
Fall 2025	Vision-assisted Transparent-aware LiDAR Mapping Main Role — Camera-LiDAR calibration (Unitree L1 LiDAR and Realsense D435i)	Autonomous (AI5003) Term Proj.

Awards

Aug. 2024	Autonomous Driving Robot Race Competition 2024-2nd (2nd Place)	KROS
Feb. 2024	2023 Winter Engineering Research Internship Presentation (Encouragement Award)	Univ. of Seoul

Skills

Programming	C++ / C, Python, MATLAB
Frameworks	ROS2 / ROS, PyTorch, OpenCV, CARLA
Tools	Git, Docker, Linux
Languages	English (TOEIC Speaking IH), Korean (Native)